



IPL Data Analysis Project

An end-to-end data analysis of **18 IPL seasons(2008–2026)**—from raw JSON to interactive Power BI dashboard.

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The Scale of the Data

Sourced from **Cricsheet.org**, this data set is one of the most comprehensive ball-by-ball cricket repositories available — requiring a custom Python parser to handle deeply nested JSON structures.

1,243

Across 18 IPL seasons

295K+

Ball-by-ball records

1,000+

One per match

18 Seasons

2008 to 2026

10 Analysis Questions

The project answers 10 key questions across five categories:



Batting

Top run-scorers, most sixes & fours



Bowling

Top wicket-takers, death-over economy



Team

Highest win count, chase vs defend



Match & Venue

Toss impact, avg score per venue



Trends

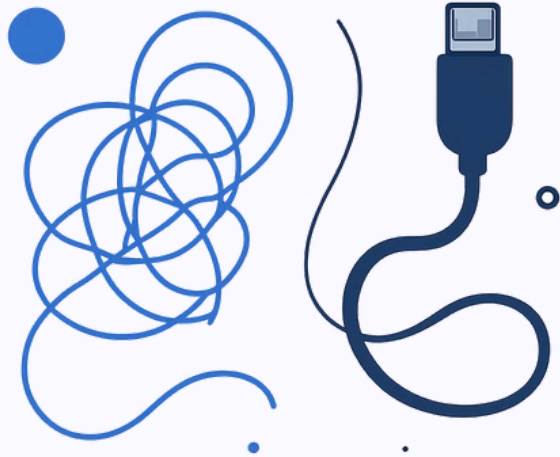
Score evolution 2008–2026, POTM awards

Data Pipeline & Methodology



Each stage transforms raw, nested match data into structured insights — culminating in an interactive two-page Power BI dashboard connected directly to MySQL.

Key Challenges & Solutions



Team Name Inconsistencies

Franchises renamed over time (e.g., Delhi Daredevils → Capitals). Fixed via SQL UPDATE across all tables.

30+ Venue Name Variations

Same stadium listed under different names. Standardized 15+ venue groups using SQL queries.

Large CSV Import Failure

29.9 MB deliveries.csv crashed MySQL Workbench. Solved using SQLAlchemy's `to_sql()` from Pandas.

Season Format Mismatch

2008 stored as '2007/08' in some records. Standardized via SQL UPDATE to restore the season slicer.

Star Schema Data Model

The cleaned data was loaded into MySQL (ipl_db) and structured as a **Star Schema** — the industry standard for analytical databases.



fact_deliveries

295,732+ rows ·
match_id, batter,
bowler, runs, is_wicket

dim_match

1,243 rows · season,
venue, toss_winner,
winner

dim_player

700+ players

dim_team

15 teams

dim_venue

64 venues

Key Findings



Batting

Virat Kohli leads with 9,000+ runs — only batter to cross this milestone. **CH Gayle** holds the sixes record.



Bowling

Bhuvneshwar Kumar tops wicket-takers. **Bumrah & Rashid Khan** are most economical in death overs (16–20).



Teams & Toss

Mumbai Indians dominate win count. Toss winners win **51.56%** of matches; teams fielding first win **53%**.



Trends

Avg first-innings score rose from ~150 (2008) to **190+** recently. Top venues: Narendra Modi Stadium & Wankhede. **AB de Villiers** leads POTM awards.

Power BI Dashboard & Skills Demonstrated

Interactive Dashboard

A two-page Power BI dashboard connected to MySQL with dynamic season slicers.

- **Page 1:** Team & Match Overview — KPIs, Win Count, Toss Winners
- **Page 2:** Player & Venue Analysis — Top Scorers, Wicket-Takers, POTM

 **DAX Measures:** Batting Avg (25.95), Economy Rate (8.15), Toss Win %, Avg Score/Match (168.57)

Skills Demonstrated



Python (Pandas, JSON)

Parsed 1,000+ files, cleaned 295K+ records



SQL & Data Modelling

Star Schema, JOIN queries, UPDATE fixes



Power BI & DAX

MySQL connection, relationships, custom measures



Data Visualization

Matplotlib charts, Power BI bar/donut/line, KPI cards